

Master Architecture Blueprint — Final Evolution Lab

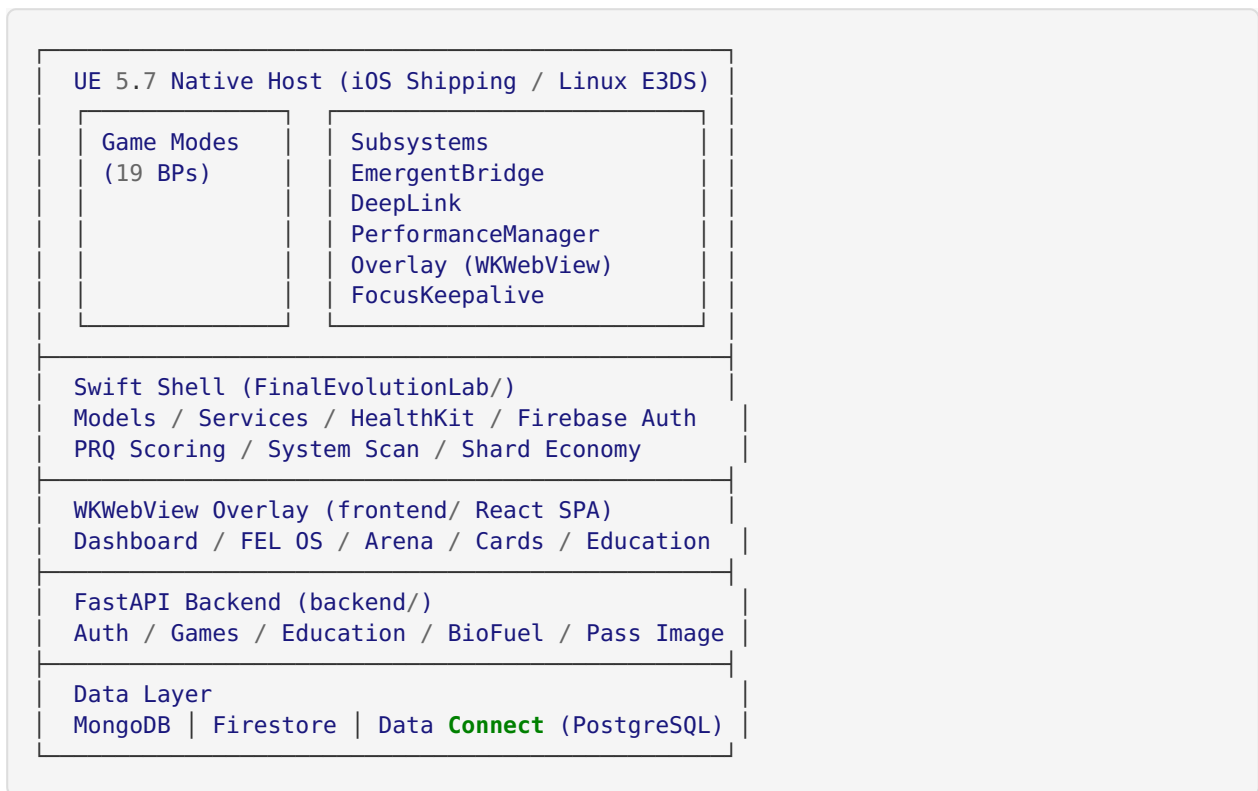
Authoritative reference for Final Evolution Lab covering the full game-mode registry (19 modes across 13 venues), map-to-mode routing, PRQ/readiness/scan truth boundaries, Creator Cards & Module Library, Shard Ledger economy, session receipt contracts, backend schemas, StoreKit/App Store compliance, Unreal Engine cooked payload pipeline, build/deploy infrastructure, and per-mode acceptance tests. Built from the `anti-gravity-fel` branch.

Goals

- Document the complete game-mode registry: 19 modes (12 production, 5 staging, 2 late additions) with per-mode map, gamemode class, binary, status, input scheme, PRQ weight, and scoring contract
 - Define the map-to-mode routing architecture across 3 layers: UE deep link subsystem, backend `ue_mode_maps.json`, and E3DS Pixel Streaming
 - Specify PRQ/readiness/scan truth boundaries: Swift HealthKit source → Firestore sync → UnrealSystemScanPayload bridge → backend weighted composite
 - Document Creator Cards & Module Library boundaries: 4-card catalog, MovementSignature physics, metricsBoost contract, PayPal purchase flow
 - Specify Shard Ledger economy: transaction types, reward formulas, coach escrow, shop catalog
 - Define session receipt data contracts: Swift GameSessionResult, backend `game_sessions` schema, XP/PRQ/Shard award formulas
 - Document backend mode schemas across MongoDB, Firestore, and Firebase Data Connect (PostgreSQL)
 - Specify StoreKit/App Store compliance requirements: prohibited distribution channels, required plist keys, export compliance
 - Define cooked payload requirements: 11 MapsToCook, descriptor-safe packaging, 3 build targets
 - Establish per-mode acceptance test matrix: launch, scoring, deep link, session persistence, PRQ impact
 - Flag critical discrepancies between registries that must be resolved before shipping
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1. System Architecture Overview

FEL is a polyglot monorepo with 5 runtime layers:



Key Source Paths

Layer	Path	Stack
UE C++ Integration	UnrealIntegration/Source/ FinalEvolutionLab/	C++/ObjC++, UE 5.7
UE Project	UnrealStarter/Basket- ballGame/	UE 5.7 Blueprints + C++
Swift App	FinalEvolutionLab/	Swift, SwiftUI, Firebase, HealthKit
React SPA	frontend/src/	React 19, Tailwind, Axios, PayPal SDK
Backend	backend/	FastAPI, MongoDB (Motor), PayPal REST
Infra	infra/	Pulumi (E3DS), UE5 config, CI scripts
Marketing Sites	sites/	React + Vite + Tailwind
Data Connect	dataconnect/	GraphQL, PostgreSQL (Cloud SQL)

2. Full Game-Mode Registry (19 Modes)

2.1 Production Modes (14)

#	mode_id	Display Name	Map	Game mode Class	Binary	Input Scheme	Sport Category	Multi-player	PRQ Weight
1	bas-ket-ball_h2h	Head to Head	Venice_Beach_Court	BP_Basketball_H2H	FEL-Basketball-H2H	charge	Basketball	real-time	1.2
2	bas-ket-ball_dunk	Dunk Contest	Venice_Beach_Court	BP_DunkContest	FEL-DunkContest	charge	Basketball	real-time	1.0
3	bas-ket-ball_3v3	3v3 Streetball	Venice_Beach_Court	BP_Basketball_3v3	FEL-Basketball-3v3	charge	Basketball	real-time	1.3
4	kar-ate_h2h	Karate · 1v1	Zen_Dojo	BP_Karate_H2H	FEL-Karate-H2H	charge	Combat	real-time	1.4
5	kar-ate_endless	Karate · Endless	Zen_Dojo	BP_Karate_Endless	FEL-Karate-Endless	charge	Combat	real-time	1.4
6	baseball	Home Run Derby	Baseball_Park	BP_Baseball	FEL-Baseball	swipe	Field	turn-Based	1.0
7	football	Kick Return	Gridiron_Stadium	BP_Football_KickReturn	FEL-Football	kick-Return	Field	turn-Based	1.5
8	soccer	Penalty Shootout	Soccer_Stadium	BP_Soccer	FEL-Soccer	penaltyKick	Field	real-time	1.1
9	golf	Closest to Pin	Links_Course	BP_Golf	FEL-Golf	swipe Golf	Precision	turn-Based	0.9

#	mode_id	Display Name	Map	Game mode Class	Binary	Input Scheme	Sport Category	Multi-player	PRQ Weight
10	tennis	Rally Ace	Tennis_Court	BP_Tennis	FEL-Tennis	rally-Ace	Precision	real-time	1.1
11	volleyball	Rally Ace	Sand_Court	BP_Volleyball	FEL-Volleyball	rally-Ace	Field	real-time	1.2
12	surfing	Surfing	Venice_Beach_Surf	BP_Surfing	FEL-Surfing	rhythmTap	Board	real-time	1.05
13	who_scene_it	Who Scene It	Neuro_Arena	BP_WhoSceneIt	FEL-Who-SceneIt	—	Academy	—	—
14	mario_party_fever	Party Mode	Venice_Beach_Court	BP_Party-Mode	FEL-Party-Mode	—	Party	—	—

2.2 Staging Modes (5)

#	mode_id	Display Name	Map	Binary	Input Scheme	Sport	Notes
15	skate-boarding	Skate-boarding	Skate_Park	FEL-Skate-boarding	rhythmTap	Board	Park Lines
16	snow-boarding	Snow-boarding	Mountain_Slope	FEL-Snow-boarding	rhythmTap	Board	Slope Control
17	gym-nastics	Gym-nastics	Training_Floor	FEL-Gym-nastics	rhythmTap	Precision	Olympic Routines
18	brain_brawl	Brain Brawl	Neuro_Arena	FEL-Brain-Brawl	—	Academy	Deep-link launch only: finale-volution://brain-brawl/launch
19	market_browse	Market Browse	Sovereign_Shop	FEL-Market	—	Commerce	3D shop browsing, no scoring

2.3 ⚠ Registry Discrepancies (Must Resolve Before Ship)

Issue	Source A	Source B	Delta
Mode count	FEL_ModeManager.production.json declares total_modes: 17	Actual entries: 19	who_scene_it + mario_party_fever added post-declaration
Swift enum	GameModeId has 17 cases	ModeManager has 19	Missing: who_scene_it, mario_party_fever, market_browse
Map routing	ue_mode_maps.json has 16 mappings	ModeManager has 19	Missing: who_scene_it, mario_party_fever, market_browse
Venue registry	FEL_VenueRegistry says total_game_modes: 16	13 venues x modes	No venue for who_scene_it, mario_party_fever, market_browse
PRQ weights	PRQScoring.swift covers 17 modes	2 production modes unweighted	who_scene_it, mario_party_fever have no PRQ weight
PRD count	PRD mentions “19 UE5 game modes”	ModeManager says 17	PRD is ahead of registry metadata

3. Map-to-Mode Routing Architecture

3.1 Three-Layer Resolution

Layer 1 — Deep Link (iOS native):

```
finalevolution://launch?map={MapToken}&mode={mode_id}
→ UFELEmergentDeepLinkSubsystem::ProcessDeepLinkUrl()
→ ParseQueryString() extracts map + mode params
→ ResolveModeToMapToken() → /Game/FEL/Maps/{MapToken}
→ OpenMapFromTokens() → UGameplayStatics::OpenLevel()
→ BroadcastMapLoaded() via EmergentBridge WebSocket
```

Layer 2 — Dashboard WebSocket (WKWebView overlay):

```
Frontend GameModesView → launchNativeMode(mode)
→ POST /api/streaming/launch-mode {mode_id}
→ Deep link to finalevolution:// scheme
→ UE subsystem processes deep link (Layer 1)
→ MapLoaded event fires on WebSocket
→ Frontend receives MapLoaded within 10s timeout
→ If timeout → system re-auth (NOT browser fallback)
```

Layer 3 — E3DS Pixel Streaming (remote):

```
backend/ue_mode_maps.json (mode_id → UE map token)
→ Pulumi e3ds/__main__.py reads game_mode_maps
→ E3DS stream config provisioned per mode
→ Travel command sent to Pixel Streaming server
→ 1920×1080, NVENC H264, 60fps, 10-30Mbps WebRTC
```

3.2 Resolution Chain Priority

1. [EmergentPlayMap] INI section (button_id → /Game/FEL/Maps/...)
2. [EmergentButtonArenaMode] INI section (button_id → arena_mode_id)
3. Hardcoded ResolveModeToMapToken() C++ fallback
4. ue_mode_maps.json for E3DS/backend routing

3.3 Venue Registry (13 Venues)

Venue	Map Path	Modes	Max Concurrent	Environment
Venice_Beach_Court	/Game/FEL/Maps/Venice_Beach_Court	basketball_h2h, basketball_dunk, basketball_3v3	4	venice_beach_sunset
Zen_Dojo	/Game/FEL/Maps/Zen_Dojo	karate_h2h, karate_endless	4	zen_dojo
Baseball_Park	/Game/FEL/Maps/Baseball_Park	baseball	2	stadium_night_game
Grid-iron_Stadium	/Game/FEL/Maps/Grid-iron_Stadium	football	2	stadium_night_game
Soccer_Stadium	/Game/FEL/Maps/Soccer_Stadium	soccer	4	classic_nba_arena
Links_Course	/Game/FEL/Maps/Links_Course	golf	4	beach_tropical
Tennis_Court	/Game/FEL/Maps/Tennis_Court	tennis	2	rooftop_cityscape
Sand_Court	/Game/FEL/Maps/Sand_Court	volleyball	4	venice_beach_sunset
Training_Floor	/Game/FEL/Maps/Training_Floor	gymnastics	2	cyberpunk_gym
Venice_Beach_Surf	/Game/FEL/Maps/Venice_Beach_Surf	surfing	2	venice_beach_sunset
Skate_Park	/Game/FEL/Maps/Skate_Park	skateboarding	4	neon_arcade
Mountain_Slope	/Game/FEL/Maps/Mountain_Slope	snowboarding	4	winter_outdoor
Neuro_Arena	/Game/FEL/Maps/Neuro_Arena	brain_brawl	8	neon_arcade

Missing from VenueRegistry: Sovereign_Shop (market_browse), Luma_Venice_Shop (in MapsToCook but no venue entry)

3.4 Sovereign Sync Protocol

- Target: M4 Pro Mac Mini
- Protocol: WSS over Cloudflare Tunnel
- Encryption: AES-256-GCM
- Signaling: Private (local network)
- UE bridge: `UFEEmergentBridgeSubsystem` with auto-reconnect, bounded outbound queue (128 messages), LAN subnet scanning for hub discovery

4. PRQ / Readiness / Scan Truth Boundaries

4.1 Source of Truth Chain

```
HealthKit (device sensors)
  [ ] Swift HealthKitService (real-time queries)
  [ ] SystemScanRecord (VitalsSnapshot + ReadinessSnapshot + AvatarPerformanceAttributes)
  [ ] Firestore users/{uid}/system_scans/{docId} (append-only archive)
  [ ] Firestore users/{uid}/avatar_performance/current (mutable latest)
  [ ] UnrealSystemScanPayload (epoch-ms JSON for UE bridge)
  [ ] UE C++ FELEmergentBridgeSubsystem (WebSocket [ ] game state)
```

4.2 Swift-Side PRQ (Neural Readiness Score)

Inputs: HRV SDNN (ms), Resting HR, Heart Rate, Active kcal, Weekly HRV average, Sleep hours

Formula:

```
hrvN = clamp(0, 1, (hrv - 18) / 72)
rhrAdj = rhr > 68 ? -12 : (rhr < 55 ? +6 : 0)
calAdj = min(12, kcal / 55)
sleepAdj = (min(1, sleep/8) - 0.5) * 18
PRQ = clamp(15, 100, 32 + hrvN*48 + calAdj + sleepAdj + rhrAdj ± random(-6, 8))
```

Grades:

Score Range	Grade	Arcade Speed	Hang Time	Recovery Mode
≥ 80	ELITE	1.15×	+0.30	No
60-79	PRIMED	1.05×	+0.15	No
40-59	READY	1.00×	0	No
< 40	RECOVERING	0.90×	-0.10	Yes

4.3 Backend PRQ (Weighted Composite)

```
{
  "strength": 0.15, "speed": 0.15, "endurance": 0.12, "agility": 0.12,
  "power": 0.12, "flexibility": 0.10, "recovery": 0.12, "mental": 0.12
}
```

- **Update trigger:** `on_session_complete`
- **Decay rate:** 0.5 per day idle

- **Boost on streak:** true

4.4 Avatar Performance Attributes (0-1 Normalized)

Attribute	Formula
explosiveness	28% tierExpl + 22% hrReserve + 18% activity + 17% rhrReserve + 15% recoveryHeadroom
neuralFocus	34% hrvVsBaseline + 28% sleepScore + 22% hrvLevel + 16% prqN + trendBoost
endurance	55% prqN + 25% activity + 20% sleep
recovery	45% recoveryHeadroom + 35% sleep + 20% hrvVsBaseline
biomechanicalEfficiency	50% hrvLevel + 30% hrvVsBaseline + 20% rhrReserve

4.5 Firestore Boundaries

- `users/{uid}/system_scans/{docId}` — Append-only. Client can create new docs but cannot update/delete historical records. One doc per sync event.
- `users/{uid}/avatar_performance/current` — Single mutable document. Latest avatar vector for UE bridge consumption. Client can read/write own doc only.

4.6 Unreal Bridge Payload Schema

```
{
  "schemaVersion": 1,
  "capturedAtEpochMs": 1716422400000,
  "vitals": {
    "heartRateBpm": 72.0,
    "restingHeartRateBpm": 55.0,
    "hrvSdnnMs": 48.0,
    "activeKcal": 320.0,
    "weeklyHrvAverageMs": 42.0,
    "sleepHoursLastNight": 7.5
  },
  "readiness": {
    "neuralReadinessScore": 78.0,
    "grade": "PRIMED",
    "hrvTrend": "IMPROVING",
    "recoveryEstimateHours": 2.0
  },
  "avatar": {
    "schemaVersion": 1,
    "updatedAtEpochMs": 1716422400000,
    "explosiveness": 0.72,
    "endurance": 0.65,
    "recovery": 0.80,
    "neuralFocus": 0.68,
    "biomechanicalEfficiency": 0.71,
    "prqScore": 78.0,
    "readinessGrade": "PRIMED",
    "speedMultiplier": 1.05,
    "hangTimeBonus": 0.15,
    "isRecoveryMode": false
  }
}
```

5. Scoring & Session Receipt Contracts

5.1 Swift Session Receipt

```
struct GameSessionResult: Codable, Sendable, Identifiable {
    let id: String
    let gameId: String           // GameModeId.rawValue
    let date: Date
    let score: Int
    let opponentScore: Int
    let shardsEarned: Int
    let prqBonus: Double
    let isMultiplayer: Bool
    let duration: Int            // seconds
    var didWin: Bool             // computed: playerScore > opponentScore
}
```

5.2 Backend Session Receipt

```
{
  "user_id": "string",
  "mode_id": "string (GameModeId rawValue)",
  "score": 0,
  "duration": 0,
  "created_at": "2026-05-22T00:00:00Z",
  "xp_awarded": "max(10, score / 5)",
  "activity_feed_entry": {
    "type": "game",
    "detail": "{mode_id}: {score} pts",
    "score": 0
  }
}
```

XP formula: $\max(10, \text{score} / 5)$ — no upper cap (risk: exploitable)

5.3 PRQ Mode Rewards

```
PRQ.modeReward(mode, won, tied, combo, criticals, scoreDifferential):
  base = won ? 2.0 : (tied ? 0.5 : 0.2)
  modeMultiplier = PRQ.modeWeight(mode) // 0.9–1.5×
  comboBonus = min(1.0, combo × 0.05)
  criticalBonus = min(0.5, criticals × 0.1)
  dominanceBonus = won ? min(0.5, scoreDiff × 0.05) : 0
  return clamp(0, 100, base × modeMultiplier + comboBonus + criticalBonus + dominance-
  Bonus)
```

5.4 Per-Mode PRQ Attribute Labels

Mode	Attribute Label	Success Chance Base
basketball_h2h / 3v3	Court IQ	0.40
basketball_dunk	Hang Time	0.45
karate_h2h / endless	Fight IQ	0.38
baseball	Bat Speed	0.35
football	Burst Speed	0.42
soccer	Shot Accuracy	0.40
golf	Swing Precision	0.30
tennis	Rally Control	0.38
volleyball	Spike Power	0.40
gymnastics	Form Score	0.35
surfing	Wave IQ	0.36
skateboarding	Line Control	0.36
snowboarding	Edge Control	0.36
brain_brawl	Cognitive Flex	0.42

6. Creator Cards & Module Library

6.1 Card Schema

```
struct CreatorCard {
  let id: String
  let creatorName: String
  let title: String
  let description: String
  let costShards: Int
  let iconName: String
  let accentColor: Color
  let metricsBoost: PerformanceMetrics // PRQ, vertical, neural drive, efficiency,
  readiness
  let movementSignature: MovementSignature {
    style: MovementStyle // explosive | vertical | fluid | neural |
    standard
    jumpApex: Double // 1.0–1.5× multiplier
    hangTimeFactor: Double // 1.0–1.8× multiplier
    firstStepBurst: Double // 1.0–1.5× multiplier
    limbEmission: Double // 0–1.0 particle intensity
    trailColor: Color
  }
}
```

6.2 Current Catalog (4 Cards)

Card ID	Creator	Cost (Shards)	PRQ Boost	Vertical Boost	Neural Drive	Move-ment Style	Anima-tion Speed
coach_v_elite	Coach V	500	+15	+20	+10	explosive	0.8×
bonds_bounce	Bonds Bounce	750	+12	+25	+8	vertical	1.0×
flight_lab	Flight Lab	600	+10	+18	+15	fluid	1.2×
neural_max	Neural Max	400	+8	+12	+25	neural	0.7×

6.3 Purchase Flow

```
User selects card in CreatorCardsView (frontend)
→ PayPal SDK creates order (backend POST /api/cards/purchase)
→ User approves on PayPal
→ Backend captures payment (paypalrestsdk)
→ MongoDB creator_cards collection updated
→ User's ownedCardIds array updated
→ metricsBoost applied to active profile
→ MovementSignature affects in-game physics
```

6.4 Avatar Skin Configuration

Cards influence avatar appearance via `AvatarSkinConfig.fromScan()` :

PRQ Range	Skin Tone	Outfit Style	Aura Color
≥ 80	elitePurple	elite	(0.6, 0.2, 1.0)
65–79	cyan	flight	(0, 0.95, 0.9)
50–64	blue	developing	(0, 0.83, 1.0)
< 50	green	standard	(0.2, 1.0, 0.4)

7. Shard Ledger Economy

7.1 Transaction Types

```
enum ShardTransaction: String, Codable {
    case workoutComplete, gameWin, gameDraw, gameLoss,
        comboBonus, criticalHit, outfitPurchase, blueprintPurchase,
        critiqueRequest, critiqueEarning, dailyBonus, streakBonus, achievementUnlock
}
```

7.2 Reward Formulas

Game Results:

Outcome	Base Shards	Bonus
Win	50	combo × 5 (if combo > 3) + criticals × 10
Draw	25	combo × 5 (if combo > 3) + criticals × 10
Loss	15	combo × 5 (if combo > 3) + criticals × 10

Workouts: exercises × base (Foundation: 10, Flight: 15, Elite: 25)

7.3 Coach Escrow System

```
struct CoachEconomy {
    var totalEarned: Int           // lifetime cleared
    var pendingEarnings: Int       // in escrow
    var clearedEarnings: Int       // released, claimable
    var escrowEntries: [EscrowEntry]
    var critiquesCompleted: Int
    var rating: Double             // weighted average
}

// Flow: critique submitted → EscrowEntry(status: .held)
//       → athlete rates → EscrowEntry(status: .released)
//       → coach claims → clearedEarnings → totalEarned
```

7.4 Shop Catalog

Category	Item	Cost (Shards)
Outfits	Neon Flux	500
Outfits	Shadow Elite	750
Outfits	Chrome V	1,000
Outfits	Gold Standard	2,000
Blueprints	Vertical Lab	300
Blueprints	Speed Matrix	300
Blueprints	Neural Recovery	200
Coaching	Form Critique	150
Coaching	Program Review	400

8. Backend Data Contracts

8.1 MongoDB Collections

Collection	Purpose	Key Fields
users	User profiles	user_id, email, prq_score, level, xp, streak, coins, avatar_config
user_sessions	Auth sessions	session_token, user_id, expires_at (ISO, 7-day TTL)
game_sessions	Game results	user_id, mode_id, score, duration, created_at
prq_metrics	PRQ history	user_id, metrics (8 weighted dimensions), timestamp
health_metrics	HealthKit syncs	user_id, hr, rhr, hrv, sleep, kcal, timestamp
education_progress	Track progress	user_id, track_id, completed_lessons[], quiz_scores{ }
bio_digital_progress	Anatomy modules	user_id, completed_modules[]
creator_cards	Card ownership	user_id, card_id, purchased_at
orders	PayPal orders	user_id, order_id, card_id, status, amount
challenges	PvP challenges	challenger_id, challenged_id, mode_id, status
streaks	Daily streaks	user_id, current_streak, last_checkin
activity_feed	Social feed	user_id, type (follow/workout/game/scan), detail, score
coach_sessions	Coaching requests	user_id, coach_id, type, status
critiques	Movement analyses	user_id, coach_id, video_id, feedback, rating
videos	Uploaded clips	user_id, url, mode_id, timestamp
workouts	Workout logs	

Collection	Purpose	Key Fields
		user_id, exercises, difficulty, completed_at
tournaments	Tournament state	tournament_id, participants[], bracket, status
enrollments	Education enrollment	user_id, track_id, enrolled_at
brain_brawl_launches	BB launch tracking	user_id, launched_at (needs TTL)

8.2 Firebase Data Connect (PostgreSQL / Cloud SQL)

```

type User @table {
  uid: String! @default(expr: "auth.uid")
  username: String
  email: String
  profilePictureUrl: String
  avatarUrl: String
  firebaseUid: String @unique
}

type Post @table {
  content: String!
  author: User!
  gameId: String
  trainingScore: Float
  clipUrl: String
  feedSource: String
  createdAt: Date! @default(expr: "request.time")
  updatedAt: Date! @default(expr: "request.time")
}

type Comment @table {
  content: String!
  post: Post!
  author: User!
  createdAt: Date! @default(expr: "request.time")
}

type PostLike @table {
  post: Post!
  user: User!
}

```

All operations require `@auth(level: USER)`. Mutations bind `firebaseUid_expr` for automatic Auth UID.

8.3 Firestore Security Rules

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /users/{userId}/system_scans/{scanId} {
      allow read: if request.auth != null && request.auth.uid == userId;
      allow create: if request.auth != null && request.auth.uid == userId;
      // No update or delete – append-only
    }
    match /users/{userId}/avatar_performance/{docId} {
      allow read, write: if request.auth != null && request.auth.uid == userId;
    }
  }
}
```

8.4 Education Tracks

Track ID	Lessons	Pass Threshold	XP/Lesson	Certificate
common_core	6	75%	50	None (P2 back-log)
stem	6	75%	50	None (P2 back-log)
kinesiology	8 + final	75% / 80%	50 + 1000	FEL-AK-{id}
brain_brawl	4 (briefings)	75%	50	None

Kinesiology Certificate — 4 Gates:

1. All 8 coursework lessons passed ($\geq 75\%$)
2. All 4 Bio-Digital anatomy modules mastered (skeletal_basics, muscular_chains, kinetic_chain_pillars, neural_priming)
3. PRQ score ≥ 80
4. Final assessment passed ($\geq 80\%$ on 10 questions)
5. → Issues FEL-AK-{id} + 1000 XP (idempotent)

9. StoreKit / App Store Compliance

9.1 Distribution Channel

APPROVED: App Store Connect / TestFlight ONLY

EXPLICITLY PROHIBITED:

- ❌ AltStore / SideStore
- ❌ OTA manifest feeds (itms-services://)
- ❌ Direct IPA install URLs
- ❌ Sideload install pages
- ❌ manifest.plist distribution

9.2 Required Info.plist Keys

Key	Value	Purpose
CFBundleIdentifier	com.finalevolutionlab.sovereign	App Store record match
CFBundleURLTypes	finalevolution://	Deep link scheme
NSHealthShareUsageDescription	(present)	HealthKit read permission
NSHealthUpdateUsageDescription	(present)	HealthKit write permission
UE_PROJECT_NAME	FinalEvolutionLab	Case-sensitive UE descriptor match

9.3 App Store Connect Requirements

- [] Export compliance declarations completed
- [] Privacy nutrition labels configured
- [] HealthKit capability declared
- [] Camera usage description (if BioFuel scanner uses camera)
- [] Encryption declarations
- [] Age rating configured

9.4 Superapp Release Metadata Contract

```
{
  "app_name": "Final Evolution Lab",
  "platform": "ios",
  "distribution_channel": "app_store_connect_testflight",
  "bundle_id": "com.finalevolutionlab.sovereign",
  "apple_app_id": "",
  "build_number": "",
  "version": "",
  "testflight_public_link": "",
  "app_store_connect_build_url": "",
  "release_notes": "",
  "created_at": ""
}
```

Generated via `./scripts/write_superapp_release_metadata.sh` → `artifacts/superapp/final-evolution-lab-ios-release.json`

10. Cooked Payload Requirements

10.1 MapsToCook (11 Maps in DefaultGame.ini)

```
[/Script/UnrealEd.ProjectPackagingSettings]
bCookAll=True
+MapsToCook=(FilePath="/Game/FEL/Maps/BaseballPark")
+MapsToCook=(FilePath="/Game/FEL/Maps/Dojo")
+MapsToCook=(FilePath="/Game/FEL/Maps/Gridiron")
+MapsToCook=(FilePath="/Game/FEL/Maps/Links")
+MapsToCook=(FilePath="/Game/FEL/Maps/Luma_Venice_Shop")
+MapsToCook=(FilePath="/Game/FEL/Maps/NeuroArena")
+MapsToCook=(FilePath="/Game/FEL/Maps/SandCourt")
+MapsToCook=(FilePath="/Game/FEL/Maps/SoccerStadium")
+MapsToCook=(FilePath="/Game/FEL/Maps/TennisCourt")
+MapsToCook=(FilePath="/Game/FEL/Maps/TrainingFloor")
+MapsToCook=(FilePath="/Game/FEL/Maps/VeniceBeach")
```

10.2 Maps NOT in MapsToCook (Staging — Will Crash if Enabled)

Map	Mode	Status	Resolution
Skate_Park	skateboarding	staging	Add when promoting to production
Mountain_Slope	snowboarding	staging	Add when promoting to production
Sovereign_Shop	market_browse	staging	Add when promoting to production
Venice_Beach_Surf	surfing	production	⚠ May share Venice-Beach token — verify

10.3 Build Targets

Target	Script	Platform	Output	Key Flags
iOS Shipping	fel_ue5_ios_shipping_package.sh	iOS	.app + .ipa in Binaries/iOS	-full-cook -shipping -export-ipa
Linux E3DS	fel_ue5_eagle3d_linux_package.sh	Linux	FEL_UE5_E3DS_Linux_Package.zip	-cook -allmaps -build -stage -pak -archive
Win64 Cook	fel_ue5_win64_cook_only.sh	Win64	artifacts/Cooked_Win64	-cook -stage -archive -skip-build -allmaps

10.4 Descriptor Safety

The iOS .app bundle MUST contain `cookeddata/` or `.pak` files. Missing cooked content causes “Failed to open descriptor file” on device launch.

Mitigation chain:

1. `fel_ue5_ios_shipping_package.sh` promotes fully staged .app from internal staging into Binaries/IOS before Xcode archive
2. Descriptor-safe IPA repacking happens BEFORE `--export-ipa` so signed .ipa is not overwritten
3. `infra/fix_ios_descriptor_path.sh` diagnostic verifies `cookeddata/.pak` presence
4. Non-iCloud project path required (xattr stripping prevents codesign failures)

10.5 Pixel Streaming 2 Configuration

```
# DefaultEngine.ini
[PixelStreaming]
Encoder=NVENC
EncoderCodec=H264
TargetBitrate=20000000

[/Script/PixelStreaming.PixelStreamingSettings]
WebRTCMaxFps=60
WebRTCMinBitrate=10000000
WebRTCMaxBitrate=30000000
AllowPixelStreamingInput=true
DefaultResolutionX=1920
DefaultResolutionY=1080

[/Script/PixelStreamingServers.PixelStreamingServers]
bUseInternalSignalling=False # E3DS manages signalling externally
```

11. Build/Deploy Checklist

11.1 Pre-Build

- [] `git checkout anti-gravity-fel && git status` — clean working tree
- [] `./infra/fel_prebuild_ci_check.sh --strict` — 6-point identifier alignment:
 1. .uproject filename = FinalEvolutionLab
 2. Target.cs class name matches
 3. DefaultGame.ini BundleIdentifier = com.finalevolutionlab.sovereign
 4. Info.plist UE_PROJECT_NAME = FinalEvolutionLab (case-sensitive)
 5. Directory paths match
 6. [Emergent] config section present
- [] Verify `backend/FEL_ModeManager.production.json` has ≥ 17 modes
- [] `./prepare_fel_full_ship.sh` — merge shipping defaults into DefaultGame.ini
- [] `./prepare_fel_emergent.sh` — merge Emergent bridge config

11.2 iOS Build

- [] Verify non-iCloud project path (suggest `~/Developer/`)
- [] `export UE_ROOT="/Users/Shared/Epic Games/UE_5.7"`
- [] `export IOS_DEVELOPMENT_TEAM="ABCDE12345"`
- [] `./fel_ue5_ios_shipping_package.sh --verify-only`

- [] `./fel_ue5_ios_shipping_package.sh --full-cook --shipping --export-ipa`
- [] Verify .app contains cookeddata/ or .pak
- [] Verify CFBundleIdentifier, HealthKit keys, URL scheme

11.3 Upload & Distribution

- [] Upload via Xcode Organizer or Transporter to App Store Connect
- [] Answer export compliance, privacy prompts
- [] Enable TestFlight
- [] `./scripts/write_superapp_release_metadata.sh` → artifacts/superapp/

11.4 E3DS Deployment

- [] `./fel_ue5_eagle3d_linux_package.sh`
- [] Set env vars: E3DS_API_KEY, E3DS_ACCOUNT_ID, FEL_BUILD_URL
- [] `./infra/deploy_e3ds.sh` — Pulumi stack provisions app, config, stream URL
- [] Verify backend/.env updated with E3DS_STREAM_URL, E3DS_IFRAME_URL, E3DS_APP_ID

11.5 Smoke Test

- [] Install from TestFlight on real device
- [] Launch app — no descriptor error, no grey screen
- [] Open WKWebView dashboard overlay
- [] Trigger `finalevolution://` deep link
- [] Exercise HealthKit permission path
- [] Verify system scan sync to Firestore
- [] Complete a game session — verify XP + Shard awards
- [] Verify E3DS Pixel Streaming connects

12. Per-Mode Acceptance Tests

12.1 Universal Test Matrix (All Production Modes)

For **each** of the 14 production modes, verify:

#	Test	Pass Criteria
1	Deep link launch	<code>finalevolution://launch?map={token}&mode={id}</code> opens correct UE map
2	MapLoaded event	WebSocket MapLoaded fires within 10s of deep link
3	Session receipt	<code>POST /api/games/session</code> accepts
4	XP award	$XP = \max(10, \text{score}/5)$ credited to user
5	Shard reward	Correct shards awarded (50 win / 25 draw / 15 loss + bonuses)
6	PRQ delta	<code>PRQ.modeReward()</code> applied with correct mode weight
7	Activity feed	Entry created with <code>type=game</code> , detail includes <code>mode_id</code> + score
8	Browser fallback	PlayableGame renders when deep link unavailable (30s timer, combos)
9	E3DS travel	Pixel Streaming travel command resolves to correct map
10	Multiplayer sync	(realtime modes only) WebSocket room join/ready/score sync works

12.2 Mode-Specific Tests

Mode	Additional Test
basketball_h2h	Charge input → shot release timing affects accuracy
basketball_dunk	Dunk contest scoring: style + height + hang time
basketball_3v3	3-player team formation, help defense logic
karate_h2h	Point sparring: strike detection + combo chains
karate_endless	Wave progression: difficulty scaling per wave
baseball	Swipe input → bat angle + timing = distance
football	Kick return: dodge mechanics + sudden death
soccer	Penalty shootout: swipe direction + power
golf	Swing arc: swipeGolf input → distance + accuracy
tennis	Rally ace: serve + volley mechanics
volleyball	Drag-to-aim spike: angle + power
surfing	Line & balance: rhythmTap timing
who_scene_it	Scene recognition quiz: timer + accuracy
mario_party_fever	Mini-game rotation: party mode round transitions

12.3 Staging Mode Gates (Before Promotion)

For each staging mode, before promotion to production:

- [] Map added to MapsToCook in DefaultGame.ini
 - [] Mode added to Swift GameModelId enum with correct rawValue
 - [] Mode added to ue_mode_maps.json
 - [] Venue entry added/updated in FEL_VenueRegistry.production.json
 - [] PRQ.modeWeight() and related functions updated
 - [] FEL_ModeManager.production.json status changed to “production”
 - [] CI mode count threshold updated if needed
 - [] Full acceptance test matrix passes
-

13. Risks and Non-Negotiable Gates

13.1 Critical (P0) — Ship Blockers

Risk	Impact	Mitigation
Registry misalignment	Deep link failures, E3DS routing errors, CI false positives	Single alignment pass across all 5 registries before any build
Descriptor safety	App crashes on launch (“Failed to open descriptor file”)	Script promotes staged .app; verify cookeddata/.pak post-build
Identifier mismatch	Build fails or app rejected by App Store	6-point CI check (<code>fel_prebuild_ci_check.sh --strict</code>)

13.2 High (P1) — Quality Gates

Risk	Impact	Mitigation
MapsToCook gaps	Staging modes crash if accidentally enabled	Feature-flag staging modes server-side; only add maps when promoting
iCloud project path	Codesign failures during iOS build	Scripts warn; developer must use ~/Developer path
E3DS Pulumi silent failure	Deployment succeeds but no stream URL	Validate ue_mode_maps.json before pulumi up; check outputs
Session token timezone	Premature/late expiry from ISO string parsing	Always compare with UTC; consider epoch-ms migration

13.3 Medium (P2) — Tech Debt

Risk	Impact	Mitigation
No XP cap per session	Exploitable high scores inflate levels	Add server-side cap (e.g., 500 XP/session)
PayPal sandbox hard-coded	No real purchases in production	Environment variable swap; add CI check for prod creds
server.py monolith (2400 LOC)	Hard to maintain/debug	Break into routers/ (auth, games, marketplace, sovereign, etc.)
No compound index on education_progress	Certificate idempotency race condition	Add (user_id, track_id) unique compound index
brain_brawl_launches no TTL	Collection grows unbounded	Add MongoDB TTL index

13.4 Non-Negotiable Gates (Must Pass Before Any Release)

1. ☒ `fel_prebuild_ci_check.sh --strict` passes (6-point alignment)
 2. ☒ Mode registry validates ≥ 17 modes in CI
 3. ☒ .app bundle contains cookeddata/ or .pak (descriptor-safe)
 4. ☒ No AltStore/SideStore/OTA/sideload references in codebase
 5. ☒ `CFBundleIdentifier` = `com.finalevolutionlab.sovereign`
 6. ☒ HealthKit usage strings present in Info.plist
 7. ☒ `finalevolution://` URL scheme registered
 8. ☒ Emergent [Emergent] section present in DefaultGame.ini
 9. ☒ All 14 production modes launch successfully via deep link
 10. ☒ Session receipts post correctly for all production modes
-

14. UE C++ Subsystem Reference

Subsystem	Type	File	Purpose
UFELEmergentBridge-Subsystem	GameInstanceSubsystem	FELEmergentBridgeSubsystem.h/.cpp	WebSocket bridge: match scores, focus keepalive, sovereign sync, hub discovery (subnet scan)
UFELEmergent-DeepLinkSubsystem	GameInstanceSubsystem	FELEmergent-DeepLinkSubsystem.h/.cpp	finalevolution:// deep link parsing, map travel, EmergentPlayMap INI resolution
UFELOverlaySubsystem	GameInstanceSubsystem	FELOverlaySubsystem.h/.cpp	WKWebView overlay lifecycle, JSON bidirectional messaging, map-loaded handshake
UFELPerformanceManagerSubsystem	GameInstanceSubsystem	FELPerformanceManagerSubsystem.h/.cpp	iOS thermal monitoring (30s poll), device tiering, dynamic resolution scaling
UFELFocusKeepaliveTickComponent	ActorComponent	FELFocusKeepaliveTickComponent.h/.cpp	Pixel Streaming 2 focus keepalive for iframe bridging
FELIOSWebOverlay	Namespace (ObjC++)	FELIOSWebOverlay.h/.mm	iOS WKWebView creation, JS eval, FEL-Bridge message handlers

Module Dependencies (FinalEvolutionLab.Build.cs)

```
PublicDependencyModuleNames: Core, CoreUObject, Engine
PrivateDependencyModuleNames: Json, RHI, RenderCore, WebSockets, Sockets
iOS-only: WebKit, MetalRHI
```

15. Design System Reference

Property	Value
Theme	Dark Clinical
Background	#050505 (default), #0F0F13 (paper), #16161A (card)
Primary	#00E5FF (cyan)
Accent Alert	#FF3366
Accent Success	#00FF9D
Heading Font	Barlow Condensed
Body Font	IBM Plex Sans
Mono Font	JetBrains Mono